

puting or calculating, but in other fields as well. That must have seemed enough for Charles Duell, the then-director of the US Patent Office, because he remarked, “Everything that can be invented has already been invented.” This is debated, but whatever the story behind it, we have an indication that the world was pretty pleased with what it had accomplished until then.



The punched-card system by Hollerith at work, counting people

1.6 The Vacuum Tube And Flip-Flops

1906

If you’ve heard of computers, you’ve heard of vacuum tubes. And if you’ve heard of those, you *should* have heard of Lee de Forest, who invented them: his first was called the Audion. De Forest helped propel the world into the Electronic Age. Where were vacuum tubes not used once they were invented? TV, radio, and later on, computers...

The site is, very interestingly, poorly designed—in fact, the homepage is near-kiddish: www.leedeforest.org. But we learn from there that during his lifetime, Lee de Forest received over 180 patents; his most significant was the three-element vacuum tube. And, about the Audion: “Earlier Thomas Edison’s electric lamp had been modified by the Englishman, Ambrose Fleming, who added a second element, called a plate, and called it the Fleming Valve. By 1906 de Forest had modified Fleming’s Valve by adding a grid to control and amplify signals, and called his device the Audion.”



The ability to control and amplifying signals does have its importance in

One of the men who thrust us into the Electronic Age